Assignment: Edge Detection using Sobel Filter

Part-A: Manual Computation

⇒ Given,

$$I = \begin{bmatrix} 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \end{bmatrix}$$

Sobel-X Kernel:

$$K = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

* 1^{st} Output location

Kernal placed on: $\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$

⇒ Element wise operation:

$$\Rightarrow (-1)(10) + (0)(10) + (1)(10) +$$
$$(-2)(10) + (0)(10) + (2)(10) + (-1)(10) + (0)(10) + (1)(10)$$

$$\Rightarrow (-10 + 0 + 10) + (-20 + 0 + 20) + (-10 + 0 + 10) = 0$$

∴ Compelete Output Matrix,

Position (1,1) will be 0

* 2nd Output location.

~~Positio~~ Position at (1,2)

$$\Rightarrow \begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} * \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow (-10+0+80)+(-20+0+160)+(-10+0+80)$

$\Rightarrow 70+140+70$

$\Rightarrow \underline{\underline{280}}$

* 3rd Output location.

Position at (1,3)

$$\Rightarrow \begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} * \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow (-10+0+80)+(-20+0+160)+(-10+0+80)$

$\Rightarrow \underline{\underline{280}}$

* 4th Output location.

Position at (2,1)

$$\Rightarrow \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix} * \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow \underline{\underline{0}}$

* 5th Output location

Position at (2,2)

$$\Rightarrow \begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} * \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow (-10+0+80)+(-20+0+160)+(-10+0+80)$

$\Rightarrow \underline{\underline{280}}$

* 6th Output location.

Position at (2, 3)

$$\Rightarrow \begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} * \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow \underline{\underline{280}}$

* 7th Output location

position at (3,1)

$$\Rightarrow \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix} * \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow \underline{\underline{0}}$

* Position at (3,2)

$$\Rightarrow \begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow \underline{\underline{280}}$

* Position at (3,3)

$$\Rightarrow \begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix} \times \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow \underline{\underline{280}}$

* Final Output Matrix.

$$\Rightarrow \begin{bmatrix} 0 & 280 & 280 \\ 0 & 280 & 280 \\ 0 & 280 & 280 \end{bmatrix}$$

* Analysis of the Output.

Edge Location

Image intensity changes from

$10 \rightarrow 80$

ie, b/w 3^{rd} and 4^{th} Columns.

Sobel-x detects the vertical edges

* Here,

the positive and negative weights of the Sobel kernel cancel out.

$(-1+0+1)+(-2+0+2)+(-1+0+1)=0$

ie, Gradient ≈ 0

ie, Output Feature Map Dimensions

For Convolution without padding

$$O = \frac{N-F}{S}+1$$

$N = 5$ (input size)
$F = 3$ (kernel size)
$S = 1$ (stride)

$$O = \frac{5-3}{1}+1 = 3$$

$\therefore$ i) Output Size = 3×3

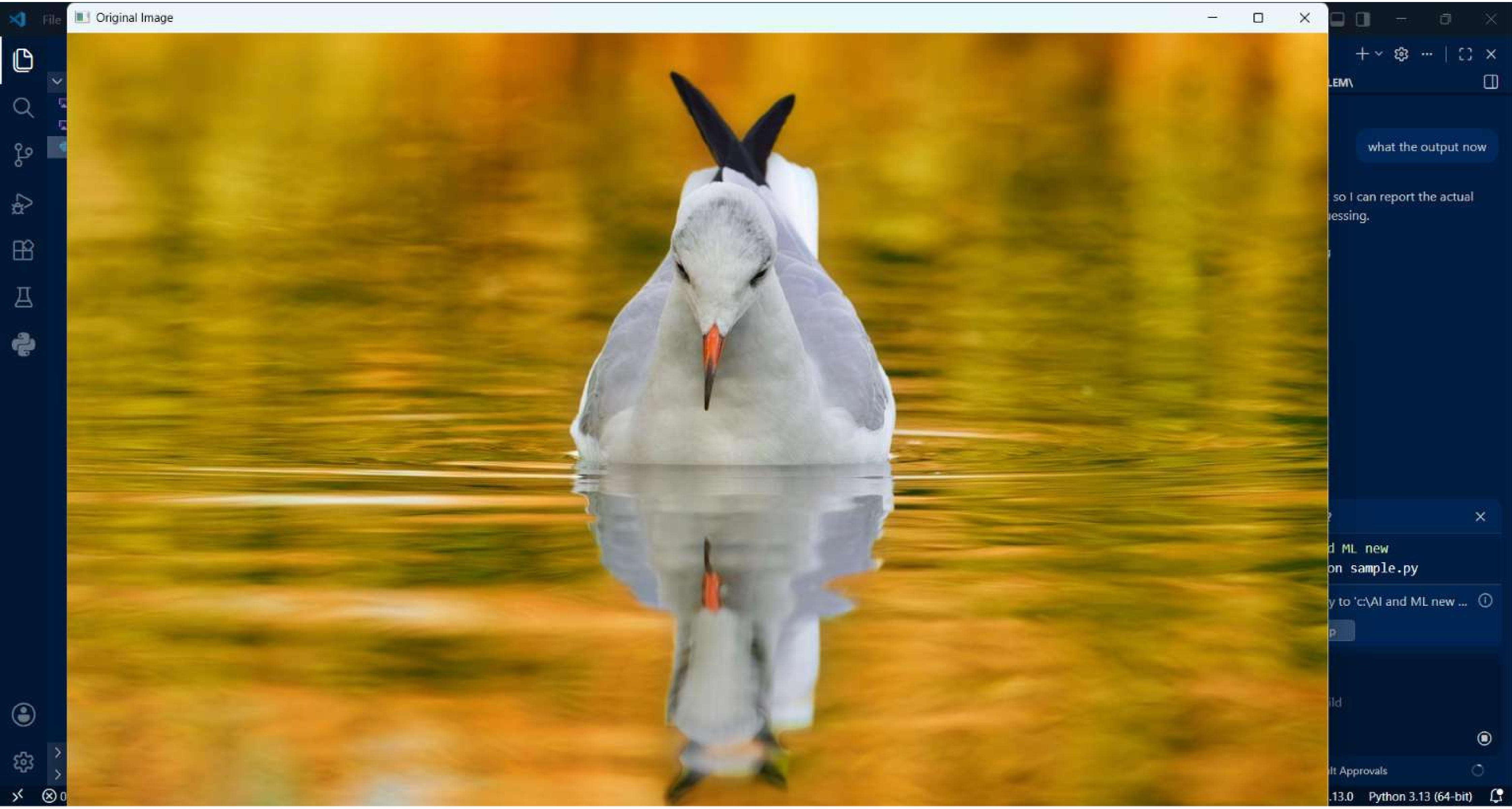
ii) First output value = 0

iii) Complete Sobel-X output:

$$\begin{bmatrix} 0 & 280 & 280 \\ 0 & 280 & 280 \\ 0 & 280 & 280 \end{bmatrix}$$

iv) Vertical edge is located where intensity changes from 10 to 80

v) Flat regions give 0 ~~resp~~ response bcz gradient is zero.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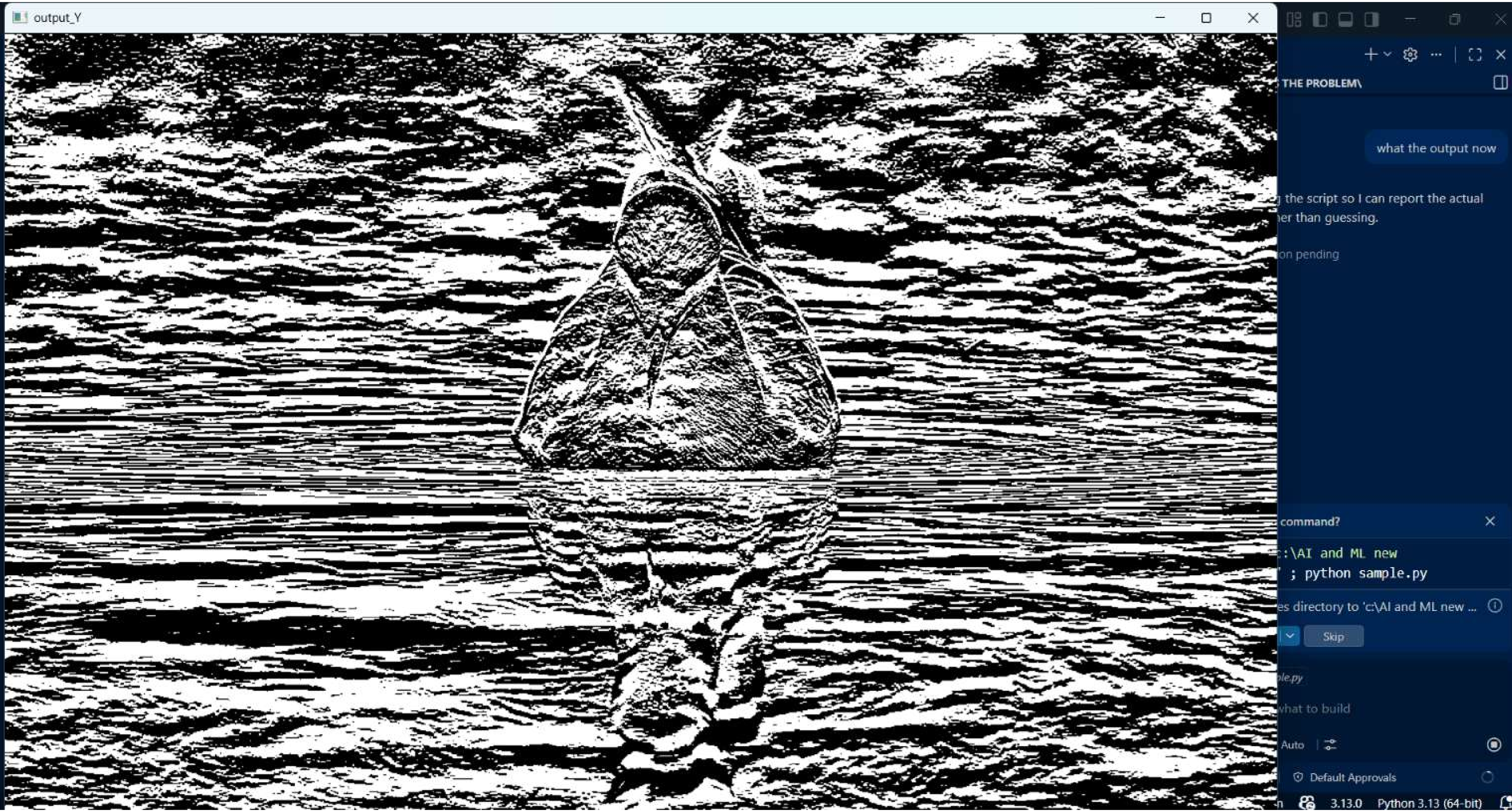
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